

DNA encoding

This article outlines a method for encoding digital data into 8-base-long DNA strands.

1 Encoding Methodology

1.1 Character to Binary Conversion

The initial step of the encoding process involves translating each character from the source data into its 8-bit ASCII binary equivalent. These individual binary representations are then concatenated in order to create a continuous binary stream, which serves as the input for the next stage.

1.2 Binary to Ternary Conversion

This long binary stream is subsequently converted into its ternary (base-3) representation. The resulting sequence of ternary digits, or trits, forms the foundational data sequence that will be encoded into DNA bases.

1.3 Sequence Partitioning

For storage, this long ternary sequence is partitioned into smaller, manageable segments, each corresponding to an 8-base-long DNA strand.

1.4 Strand Structure and Error Correction

The information within each 8-base-long DNA strand is structured to include indexing trits, data trits, and an error correction trit. The value for this error correction trit is calculated for each segment by summing the corresponding source trits (both index and data) and taking the result modulo 3. This integrated approach aims to create an efficient and reliable method for DNA-based data storage.

1.5 Base Conversion Rules

The conversion from these ternary digits (trits) to DNA bases is specifically designed to prevent the formation of homopolymers—consecutive runs of the same base. This measure significantly reduces the likelihood of synthesis and

sequencing errors. To ensure consistent encoding across all sequences, the first base of every DNA strand is processed using a fixed initial condition: the preceding base is always treated as 'G'. Any necessary padding should appear at the beginning.

Table 1: DNA Base Conversion Rules from Ternary Digits

Previous Base	Next Base for Ternary Digit		
	0	1	2
A	T	C	G
T	A	C	G
C	A	T	G
G	A	T	C

1.6 Decoding

The retrieval of the original information is achieved through DNA decoding, a process that directly reverses these encoding steps to reconstruct the initial binary data.

2 Conclusion

The described DNA encoding method offers a structured approach to data storage that directly addresses the challenge of homopolymer-induced errors. It achieves a great balance between efficiency and data integrity for storing with DNA strands that are 8 bases in length. This methodology not only ensures data integrity but also provides a straightforward and reversible pathway, making both the DNA encoding and DNA decoding processes highly efficient.

In the context of decoding from raw FASTQ sequencing data, the adapter sequence serves as a positional marker to identify the start of the payload data. Once the payloads are extracted, the specific range of valid indexes must be determined before the consensus mechanism is applied. Within this defined range, the highest frequency valid 8-mer DNA strand is identified for each index to construct a robust consensus sequence for the decoding phase.